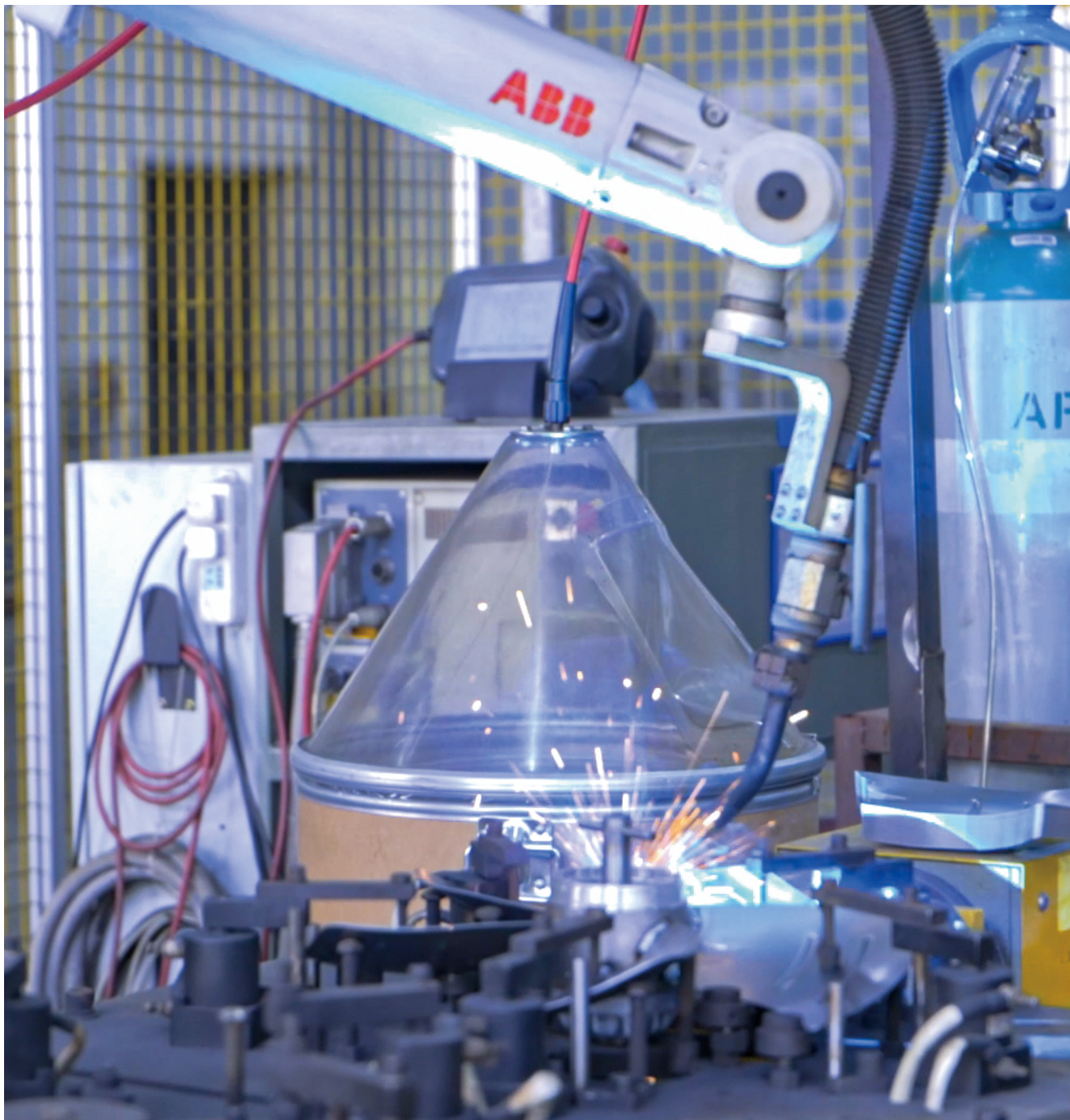


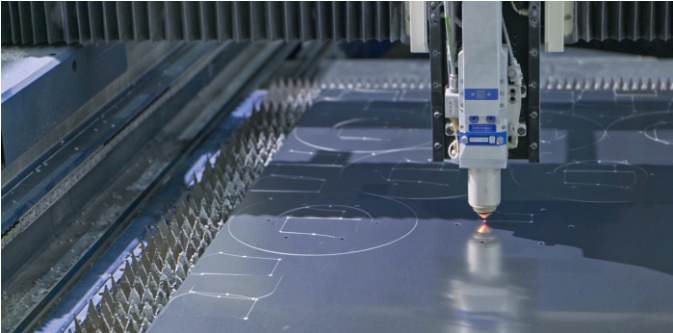
Factory Workshop



Seemtek currently has various fully automated production lines in use, producing millions of high-performance axial fans and centrifugal fans every year. These production lines ensure the highest quality and efficiency, meeting the rigorous demands of our global customers. With the expertise and dedication of our skilled workforce and advanced manufacturing techniques allows us to deliver superior products that set industry standards for performance and reliability.

20+
Years of
Experience
in manufacturing

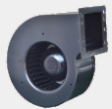
Millions of
High-
Performance
Axial
Centrifugal Fans



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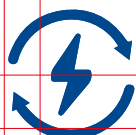
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Principles of EC Fan Model Selection

When selecting an EC fan model, some key factors have to be considered to ensure the chosen fan meet the specific needs of the application while delivering optimal performance and energy savings. The principles below can serve as a guide for making informed decisions in EC fan model selection.



Efficiency

The primary principle guiding the selection of an EC fan model is efficiency. The efficiency of an EC fan is determined by its ability to convert electrical energy into mechanical energy while minimizing losses due to heat dissipation. When selecting an EC fan model, it is important to consider its efficiency rating, typically expressed as a percentage. Higher efficiency ratings indicate a more energy-efficient fan, resulting in reduced power consumption and operating costs.



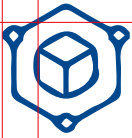
Airflow and Pressure

Another crucial principle in EC fan model selection is matching the airflow and pressure requirements of the application, known as the operating conditions. In order to achieve a certain airflow, the corresponding system resistance needs to be overcome. The airflow requirement is determined by factors such as the size of the space being ventilated, the desired air exchange rate, and any specific environmental conditions that need to be addressed (e.g., temperature or humidity control). To ensure optimal performance, it is essential to select an EC fan model that can deliver the required airflow while operating within its design parameters. This includes considering the fan's maximum airflow capacity and the ability to adjust the fan speed to meet varying airflow demands.



Noise Level

Noise reduction is an important consideration in many applications, particularly in environments where low noise levels are critical. When selecting an EC fan model, it is important to consider its noise level rating, typically expressed in decibels (dB). Lower noise level ratings indicate a quieter fan, ensuring a more comfortable and productive environment.



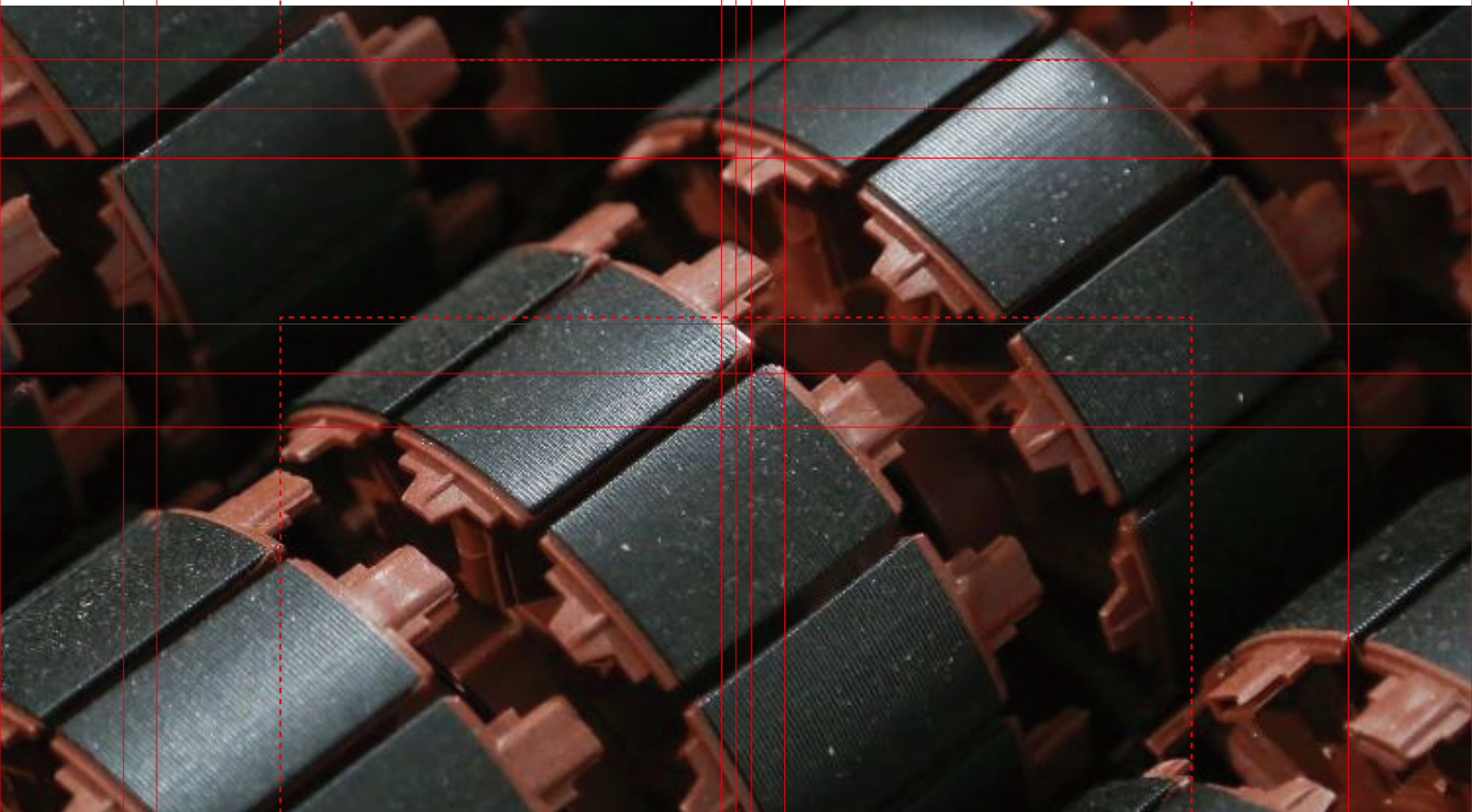
Controls and Integration

The ability to integrate the fan into a broader control system is another principle to consider in EC fan model selection. EC fans offer a range of control options, such as variable speed control and the ability to communicate with building management systems. These features allow for precise control over the fan's operation, enabling energy savings through demand-based ventilation. When selecting an EC fan model, it is important to ensure compatibility with the desired control system and assess the available control options to maximize energy efficiency.



Maintenance and Reliability

Lastly, the principle of maintenance and reliability should be considered when selecting an EC fan model. It is essential to choose a fan model that is designed for durability and reliability, reducing the need for frequent maintenance or replacement. Assessing the service life and reliability test that has been done can help ensure a stable operation of the chosen EC fan model.



Part Numbering System

B 3 E - 2A 190 - S0 - 00
 | | | | | | |
1 2 3 4 5 6 7

1: Fan Type

B: Backward Curved Centrifugal Fan
 K: Centrifugal Fan with Bracket
 F: Forward Curved Centrifugal Fan
 D: Dual Inlet Blower Fan
 S: Single Inlet Blower Fan
 A: Axial Fan
 G: Axial Fan with Guard Grille
 W: Axial Fan with Wall Ring
 M: Motor

2: Number of Cores(EC) / Number of Poles(AC)

EC: 1-core and 3-core
 AC: 2, 4, 6, 8 and 12 poles

3: Type of Motor

E: EC brushless motor
 D: DC brushless motor
 A: AC single phase induction motor

4: Input Voltage

12: 12 VDC
 24: 24 VDC
 48: 48 VDC
 85: 85 VDC
 1D: 110 VDC
 3D: 310 VDC
 1A: 115 VAC
 2A: 230 VAC
 3A: 380 VAC

5: Diameter

190: impeller diameter in mm

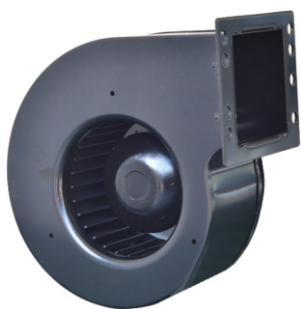
6: Key for Electrical Design

7: Key for Mechanical Design

01

EC Single Inlet Blower Fan

EC Single Inlet Blower Fan



Impeller Material: PA66

Direction of Rotation: Clockwise, seen on rotor

Speed Control: 0-10VDC/PWM

Type of Protection: IP44

Operation Mode: S1

Bearing: Maintenance-free ball bearings

Ambient Temperature: - 25 °C ~ +60 °C

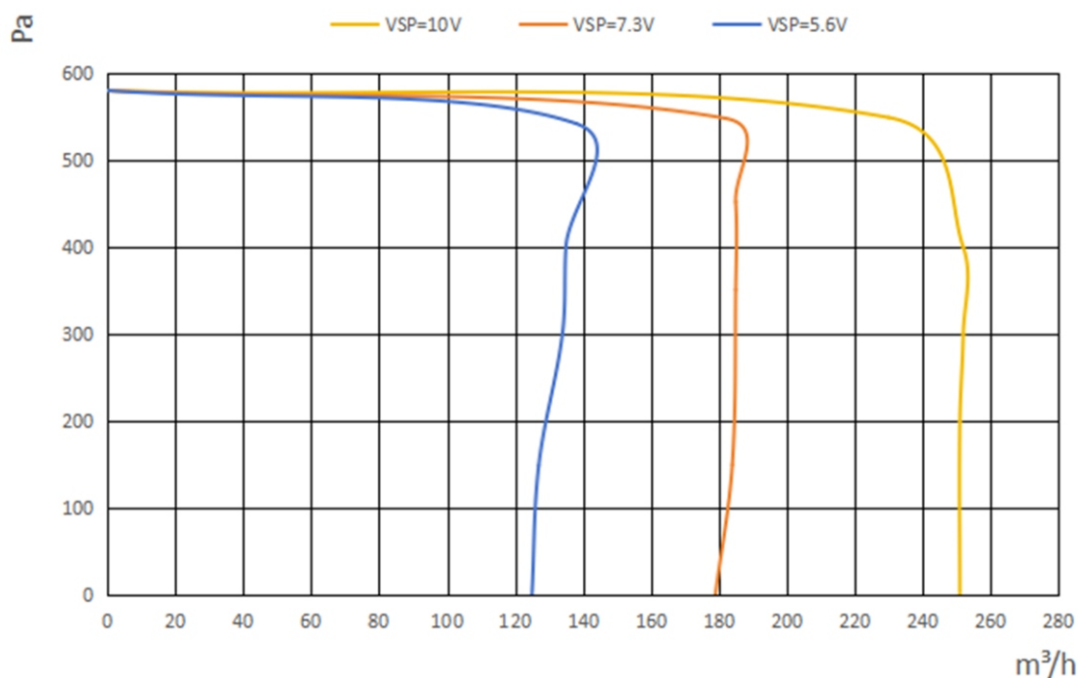
Frequency: 50/60 Hz



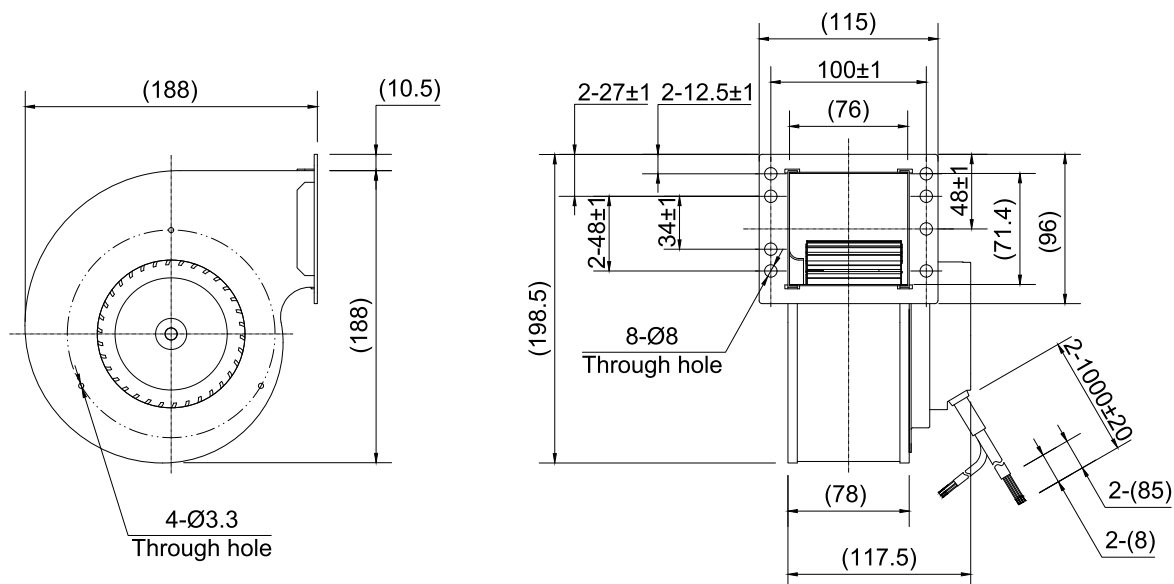
Technical Parameters

Model No.	Rated Voltage VAC	Rated Current A	Rated Power W	Rated Speed RPM	Air Flow m ³ /h	Air Pressure Pa	Insulation	Noise Level dBA
S3E-2A120-A7-01	220	0.25	35	1850	250	580	Class B	44

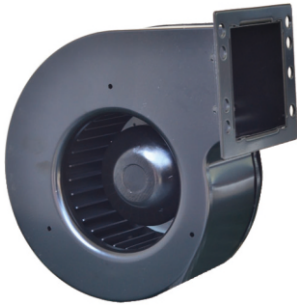
Performance Curve



Outline Drawing



EC Single Inlet Blower Fan



Impeller Material: PA66

Direction of Rotation: Clockwise, seen on rotor

Speed Control: 0-10VDC/PWM

Type of Protection: IP44

Operation Mode: S1

Bearing: Maintenance-free ball bearings

Ambient Temperature: - 25 °C ~ +60 °C

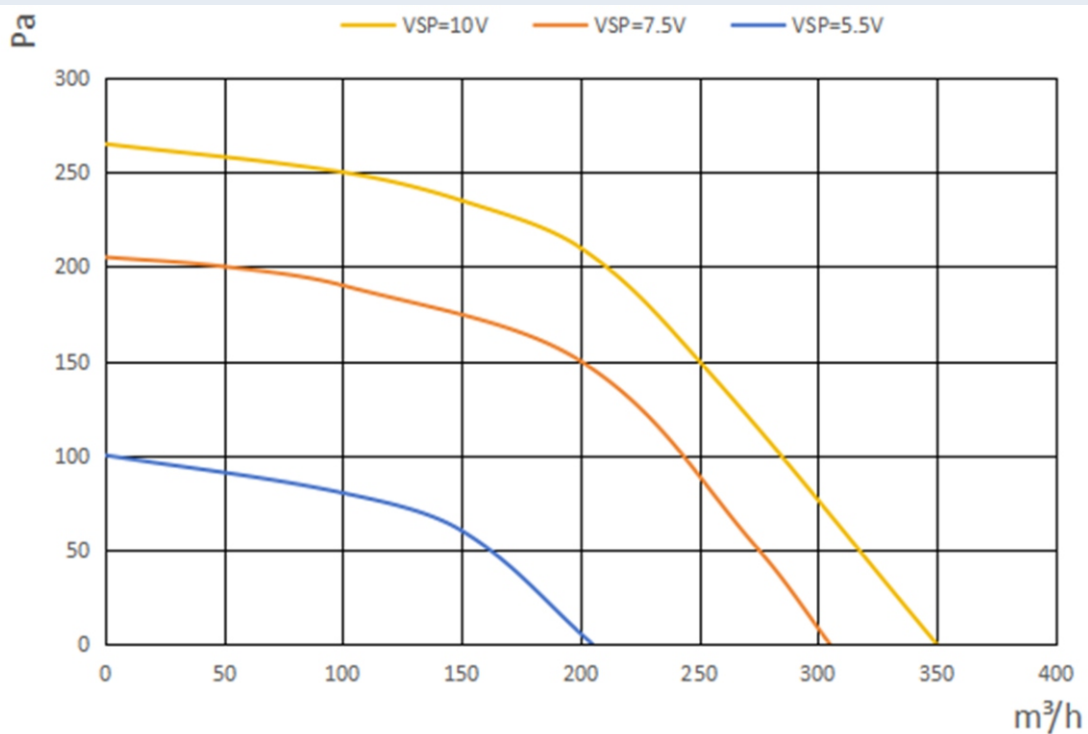
Frequency: 50/60 Hz



Technical Parameters

Model No.	Rated Voltage VAC	Rated Current A	Rated Power W	Rated Speed RPM	Air Flow m³/h	Air Pressure Pa	Insulation	Noise Level dBA
S3E-2A120-A7-02	220	0.50	70	2400	350	265	Class B	66

Performance Curve



Outline Drawing

